

AGROPORTAL: AN INTELLIGENT AI-POWERED PLATFORM FOR PRECISION AGRICULTURE AND FARMER ASSISTANCE

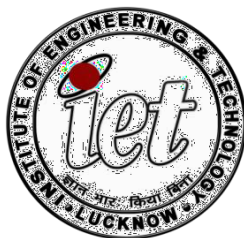
A Project Report Submitted in Partial Fulfillment of the Requirements
for the Degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE &
ENGINEERING
(ICS-851)

Bachelor of Technology
in
Computer Science and Engineering

Submitted by

2200520100011 Amit Pandey
2200520100015 Anshul Pandey
2200520100025 Durgesh Agrahari

Under the guidance of
Dr. Pawan Kumar Tiwari
Ms. Deepa Verma



Department of Computer Science and Engineering
INSTITUTE OF ENGINEERING AND TECHNOLOGY
Dr. A.P.J. Abdul Kalam Technical University Uttar Pradesh

2025-2026

DECLARATION

We affirm that this submission is entirely our own work and, to the best of our knowledge and belief, it does not include any material previously published or authored by someone else, nor does it contain any content that has been accepted for the award of any degree or diploma at a university or other higher education institutions, except where proper acknowledgment is made in the text. This project has not been submitted by us to any other institution for the fulfillment of any other degree requirements. The findings and technical implementations documented herein represent a comprehensive effort to bridge the technological divide within the agricultural sector through data-driven methodologies.

Submitted by:

Amit Pandey (2200520100011)

Anshul Pandey (2200520100015)

Durgesh Agrahari (2200520100025)

CERTIFICATE

This document certifies that the project work carried out by Amit Pandey, Anshul Pandey, and Durgesh Agrahari, under my guidance and supervision at the Department of Computer Science and Engineering, Institute of Engineering and Technology, Lucknow, is detailed in the project report titled **"AGROPORAL: An Intelligent AI-Powered Platform for Precision Agriculture and Farmer Assistance"**. This report was submitted as part of the requirements for obtaining a Bachelor of Technology degree in Computer Science and Engineering. Furthermore, to the best of my knowledge, this project has not been submitted to any other institution for the purpose of earning any other degrees.

(Dr. Pawan Kumar Tiwari)

Department of Computer Science and Engineering
Institute of Engineering and Technology, Lucknow

(Ms. Deepa Verma)

Department of Computer Science and Engineering
Institute of Engineering and Technology, Lucknow

ACKNOWLEDGEMENT

The culmination of this research project was made possible through the collective support and intellectual guidance provided by the faculty and peers at the Institute of Engineering and Technology. We wish to recognize and thank everyone who offered their support throughout our study. We are profoundly thankful to the Almighty for the grace and blessings that allowed us to finish this project.

Our supervisors, **Dr. Pawan Kumar Tiwari** and **Ms. Deepa Verma**, deserve our sincere gratitude for their constant assistance and technical direction. Their vast knowledge and patience have been essential to the success of this B.Tech project. We are also appreciative of the project coordinator for providing us the opportunity and resources to carry out our study. Additionally, we extend our thanks to the Head of the CSE Department for providing the necessary tools and administrative assistance that facilitated a smooth research process.

ABSTRACT

The Agricultural Portal is an intelligent, data-driven web platform designed to support farmers in making accurate and timely decisions by integrating key modules such as crop recommendation, fertilizer recommendation, real-time weather forecasting, and agriculture-related news updates. Using machine learning algorithms and reliable environmental datasets, the system provides personalized insights based on soil characteristics, climatic conditions, and user inputs, helping farmers optimize crop selection, resource management, and overall production planning. This paper outlines the complete development process of the portal, including system architecture, model selection, dataset preparation, and implementation of each functional component, with a strong focus on creating a clean, intuitive, and accessible user interface suitable even for those with minimal technical knowledge. The platform's modular and scalable design ensures efficient performance, easy maintenance, and potential for future enhancements. Beyond supporting day-to-day farming decisions, the portal promotes sustainable and data-driven agricultural practices by offering predictive analytics, environmental insights, and timely updates that help reduce uncertainty and mitigate risks. As digital transformation continues to reshape the agricultural sector, this Agricultural Portal serves as an essential tool for evolving traditional farming into a more efficient, intelligent, and resilient system.

CONTENTS

Section	Page No.
DECLARATION	i
CERTIFICATE	ii
ACKNOWLEDGEMENT	iii
ABSTRACT	iv
CHAPTER 1: INTRODUCTION	1
1.1 Problem Statement	2
1.2 Motivation	3
1.3 Objective	4
CHAPTER 2: LITERATURE REVIEW	5
CHAPTER 3: METHODOLOGY	8
3.1 System Architecture	8
3.2 Proposed Methodology	9
3.3 Algorithm Used	10
3.4 Development Tools and Frameworks	18
3.5 Implementation	20
CHAPTER 4: RESULTS & DISCUSSION	22
4.1 Training Details	22
4.2 Result	22
CHAPTER 5: CONCLUSION	29
5.1 Conclusion	29
5.2 Future Work	29
REFERENCES	30

FIGURES

Figures No.	Figure Captions	Page No.
Figures 4.1	Crop Recommendation	24
Figures 4.2	Fertilizer Recommendation	25
Figures 4.3	Crop Prediction	26
Figures 4.4	Yield Prediction	27
Figures 4.5	Weather Forecast	28

CHAPTER 1

INTRODUCTION

The modern agricultural landscape is undergoing a fundamental transition as digital and AI-based technologies are quickly adopted to improve productivity, efficiency and sustainability. Agriculture remains one of the most essential sectors of the global economy mainly in developing nations like India where a large portion approx. 60% of the population depend on farming for food. The agricultural sector continues to face major challenges such as unpredictable climate changes, irregular rainfall patterns, soil degradation due to widely chemical usage and fragmented supply chains that often disadvantage farmers cautiously. Traditional farming methods are no longer sufficient to deal with the fast changing conditions of modern agriculture.

Predicting agricultural result has become difficult because multiple factors such as soil nutrient, humidity, rainfall, temperature and market demand directly effect crop production. Unsuitable crop selection and fertilizer usage produce less yield and losses like income of farmers. To overcome these problems the integration of AI and ML provides a more accurate and data-driven approach to agricultural management. Machine learning algorithms can analyze large datasets, identify patterns, and generate intelligent predictions that help farmers to make decisions regarding farming activities.

Traditional farming methods are being modernized by AGROPORTAL which is a smart, AI driven digital platform built for precision agriculture. A variety of essential farming services are added in the single web-based platform. Through it, users can access crop and fertilizer suggestions, harvest yield predictions, rainfall studies, weather reports, market trading, farming news and help from an AI chatbot. To give highly accurate advice, soil factors including Nitrogen (N), Phosphorus (P), Potassium (K), moisture, temperature, and rain levels are analysis by the system using machine learning models like Random Forest, Decision Tree and Naive Bayes. Live weather updates are added into the platform by using third party that is the Open Weather Map API, which allows farmers to easily plan out their watering, harvesting and fertilizing schedules.

1.1 Problem Statement

Farmers today face a systemic lack of transparency and reliable information, leading to suboptimal decision-making. Important factors such as precise crop prediction, crop recommendation, rainfall prediction, fertilizer recommendation and yield prediction. The dependency on old methods carry to decrease productivity, improper usage of fertilizer, and avoidable crop losses.

Besides, there is a different lack of integrated digital tools. While individual solutions exist for weather or soil testing, there is no single, unified web-based platform that combines soil-based crop recommendations, fertilizer guidance, yield prediction, and real-time weather updates. This division confuses the farmer's journey, forcing them to search for various inputs across disorderly platforms. Also, the lack of direct market access creates a dependency on middlemen, which optimises income for intermediaries while confusing the financial stability of the farmer.

Problem Category	Specific Challenge	Impact
Technical Gap	Fragmented AI solutions	Incomplete tools for decision-making
Environmental	Climate variability	Unpredictable rainfall and temperature shifts
Operational	Reliance on traditional experience	Inaccurate assessment of soil health
Economic	Market middlemen dependency	Unfair pricing and reduced farmer income
Digital Divide	No unified portal	Farmers must consult multiple platforms

1.2 Motivation

The motivation behind AGROPORTAL is fixed in the belief that technology should empower the agricultural sector, which forms the backbone of the economy and supports millions of livings. In many rural regions, farmers still rely on old farming practices, old information, false information and manual decision-making methods for crop usage, fertilizer usage, and irrigation planning. The lack of access of accurate agricultural information often leads to low production, soil degradation, improper resource utilization, also doesn't have the weather information and income losses. AGROPORTAL was developed with the aim of reducing this information gap by providing information to farmers with a smart, reliable, and data-driven agricultural guidance platform.

With the increasing of availability of internet connectivity and digital services in rural areas, there is a important opportunity to introduce smart agricultural solutions that are simple, accessible, and practical for everyday farming activities. The project is motivated by the growing need for precision agriculture, where decisions are based on scientific analysis rather than blind assumptions. By integrating Machine Learning algorithms such as Random Forest, Decision Tree helps the farmers receive accurate crop recommendations, fertilizer suggestions, crop yield predictions, and rainfall analysis based on soil, third party platform and environmental conditions.

Another major motivation behind this project is to support sustainable farming and efficient usage of resource management. Extra use of fertilizer and poor crop planning reduce soil fertility and also increase production costs for farmers. By intelligently prediction systems and soil-based recommendations, AGROPORTAL aims to optimize agricultural resources, improve crop productivity, and minimize environmental damage caused by improper farming practices like fertilizer, ploughing etc. Additionally, the integration of real-time third party weather forecasting help farmers to better prepare for changing climatic conditions and reduce risks linked with unpredictable rainfall and temperature change.

The project is also motivated by the economic challenges faced by farmers due to irregular supply chains and dependency on middlemen for selling agricultural produce. AGROPORTAL addresses this issue by providing a digital marketplace where farmers can directly connect with customers and sell their products. The inclusion of an AI powered chatbot, agricultural news feeds and government advisory updates helps farmers stay informed about modern techniques, schemes and practices. Overall, the motivation behind AGROPORTAL is to create a smart platform that promotes digital transformation, workable farming, and improved quality of life for farming related communities.

1.3 Objective

The primary objective of this project is to develop an good, smart, and unified Agricultural Web-Oriented Portal that mixes multiple decision to support tools into one user-friendly platform. The specific goals are as follows:

- **To Develop the Intelligent Recommendation Model using machine learning**
- **To Optimize the Resource through fertilizer recommendation**
- **To Increase Prediction Ability using crop yield and rainfall forecasting**
- **To Link the Information Divide with real-time agricultural updates**
- **To Shape a Scalable MERN Platform integrated with machine learning models**

CHAPTER 2

LITERATURE REVIEW

AI-driven crop disease monitoring systems are emerging as one of the most effective solutions to today's modern farming challenges, as discovered by recent studies. Past research has pointed to the time, subjectivity, and the experience that involved in old visual assessment by farmers as some of the drawbacks. To solve these challenges, many recent studies support the application of deep learning, particularly Convolutional Neural Networks, to automatically recognize disease patterns from leaf images. Patel & Singh (2022) had proposed CNN based model which was able to extract the fine-grained features like colour variations, texture, shapes of the lesions etc. which yields much higher accuracy than traditional machine learning models. Several other works also highlight the need to enhance the strength of the model in varying lighting and environmental conditions by using image datasets, data augmentation, and training optimizers. Research are indicates that combining AI models with cloud systems enables farmers to get real-time disease forecasts in the field. From the literature, it can be observed that AI-based disease detection systems provide quicker diagnosis, minimize crop losses and aid in creating an effective and scalable smart agriculture system all of which align with the result of the proposed project [1].

Agriculture in India faces several difficulties such as outdated farming methods, climate uncertainty, and limited access to modern technology. Recent studies have shown that Artificial Intelligence and Machine Learning can improve agricultural production through crop prediction, disease detection, and yield forecasting. Researchers such as Rahman, Shamrat, Tasnim, Roy, and Hossain (2019) highlighted the use of supervised Machine Learning algorithms in prediction systems and decision-making models [2].

Many existing agricultural systems provide separate services such as weather forecasting, crop recommendation or online trading but they lack in added user-friendly platform for farmers. To overcome these problem the proposed Agricultural Assistance & Trading platform combines Machine Learning models, AI based tools, weather forecasting, agricultural news and crop trading features into a one platform. These system helps farmers to take the informed decisions, improve production and access modern agricultural technologies more efficiently.

Finally, the Agricultural Assistance & Trading Portal establishes how technologies such as Machine Learning, Artificial Intelligence, third party APIs, and full-stack web development can be effectively applied in the agricultural sector. The system provides decision-support tools, secure digital trading, and real-time agricultural assistance through an platform. By improving accessibility to smart farming technologies and reducing the gap between farmers and consumers, the project contributes toward efficient, maintainable and technology driven agriculture.

Existing research in smart agriculture focus the importance of Machine Learning and data driven systems in farmers decision-making processes. Various studies have used parameters such as temperature, rainfall, soil type, season, and cultivation area for crop and yield prediction. Researchers have applied algorithms including Artificial Neural Networks, Regression Models, K-Nearest Neighbors, Decision Trees, Random Forests and Stacking Regression increase prediction accuracy based on the training data and agricultural planning.

Several studies also focus on fertiliser recommendation, weather forecasting and crop suitability based on the soil and climate data. Researchers have focus on the use of association rule of mining and predictive analysis to find out the relationships between agricultural parameters and farming outcomes. Liu, Zhang, and Wang (2025) discussed the recent advancements of technology and the challenges faces in Machine Learning based on crop yield prediction systems and focus on the growing of agriculture technologies in improving production and sustainability [3].

These research contributes toward the integration of Machine Learning models with web based platforms. which significantly improve agricultural accuracy and farmer support systems. Inspired from these advancements, the proposed Agricultural Assistance & Trading Platform combines crop prediction, yield prediction, weather forecasting, fertilizer recommendation, chatbot assistance and online crop trading into a single platform. The system aims to provide real-time recommendations, decision support and improved access of modern agricultural technologies for farmers and customers.

Recent research focus on the importance of Artificial Intelligence, Internet of Things and precision agriculture in achieving maintainable and climate based farming. As the technologies increases such as Machine Learning, deep learning, remote sensing are widely used for soil analysis, irrigation optimization, crop recommendation and disease detection. Precision farming tools like GPS based systems, Variable Rate Technology help to improve resource utilization of farmer, reduce wastage of resources, increase the income sources and increase agricultural production. Manjula and Djodiltachoumy (2017) proposed a crop yield prediction model that demonstrated the effective of computational techniques in improving agricultural planning and Production [4].

Many existing agricultural systems that focus only on individual services, lack of various information in the single platform as well as user-friendly seamless platform. To overcome these problems, the proposed Agricultural Platform & Trading Portal combines with Machine Learning models, AI chatbot support, crop prediction, weather forecasting, fertilizer recommendation, and online crop trading within a single platform that helps a lot to customer and farmer. The system aims to provide intelligent, accessible, and technology driven agricultural support that promotes sustainable farming and improves farmer productivity.

Agriculture is one of the most important sectors on which many people depend for their income growth and food availability, but it faces many challenges due to rapid climate change, unpredictable weather conditions, unpredictable crop prediction and limited agricultural resources. Recent reserach points the rapid use of Artificial Intelligence and Machine Learning in improving agricultural production and many decision making. Techniques such as supervised learning, unsupervised learning, reinforcement learning, and Long Short-Term Memory models are widely used for crop yield prediction, weather forecasting, irrigation planning and resource optimization. Karishma Kumari, Ali Mirzakhani Nafchi, Salman Mirzaee and Ahmed Abdalla (2025) discussed the use of

AI driven technologies in achieving climate smart and sustainable agriculture through intelligent prediction and analysis systems [5].

In spite of these advancements many existing agricultural solutions that focus only on specific functionalities and do not provide a centralized platform for multiple functionalities like crop prediction, yield prediction, rainfall prediction, crop recommendation, fertilizer recommendation. To address this problems, the proposed Agricultural platform and Trading Portal includes crop prediction, fertilizer recommendation, yield estimation, weather forecasting, chatbot assistance, agricultural news and online crop trading into a single platform. The system provides a intelligent and user-friendly solution that provide modern farming practices, improves decision-making and promotes sustainable agriculture through technology driven platform.

CHAPTER 3

METHODOLOGY

The methodology plans the step-by-step process followed to develop the Agricultural Web Portal. The key goal is to create a simple, useful and reliable system that tries to support farmers with accurate data, important information and some basic prediction features. It describes the approach used to convert raw environmental data into worthwhile agricultural intelligence. Our goal is to create a system that supports farmers with right predictions and all-in-one market access.

3.1 System Architecture

AGROPORTAL is built on the MERN tech stack with a Python using ML. The system used to follow client-server architecture where modern frontend, backend, database, and machine learning technologies work together to provide agricultural services. The frontend is developed using React.js with Vite to create a fast, responsive, and component-based user interface for farmers, customers, and admin. The backend uses Node.js and Express.js to manage REST APIs, authentication, routing, and communication between different system modules. MongoDB with Mongoose ODM is used as the database layer to store user records, crop listings, orders, and other data in a flexible NoSQL structure.

The machine learning layer is implemented using Python and scikit-learn libraries to perform various functionality. These ML models are connected with the backend through Node.js `child_process` integration, which enables real-time prediction generation. External APIs such as OpenWeatherMap, NewsAPI and Groq are integrated to provide weather forecasting, agricultural news updates, and AI-powered chatbot assistance. The overall architecture is going to ensure scalability, modularity, secure data management, effective processing and smooth interaction between users and prediction services.

Layer	Technology	Purpose
Frontend	React.js (Vite)	Provides responsive Single Page Application (SPA) interface and dynamic UI components
Backend	Node.js + Express.js	Handles REST APIs, routing, authentication, and server-side logic
Database	MongoDB (Mongoose ODM)	Stores users, products, orders, predictions, and application data
ML Engine	Python + scikit-learn	ML inference scripts invoked via <code>child_process</code>
External APIs	OpenWeatherMap, NewsAPI, Groq	Provides weather updates, agricultural news, and AI chatbot functionality

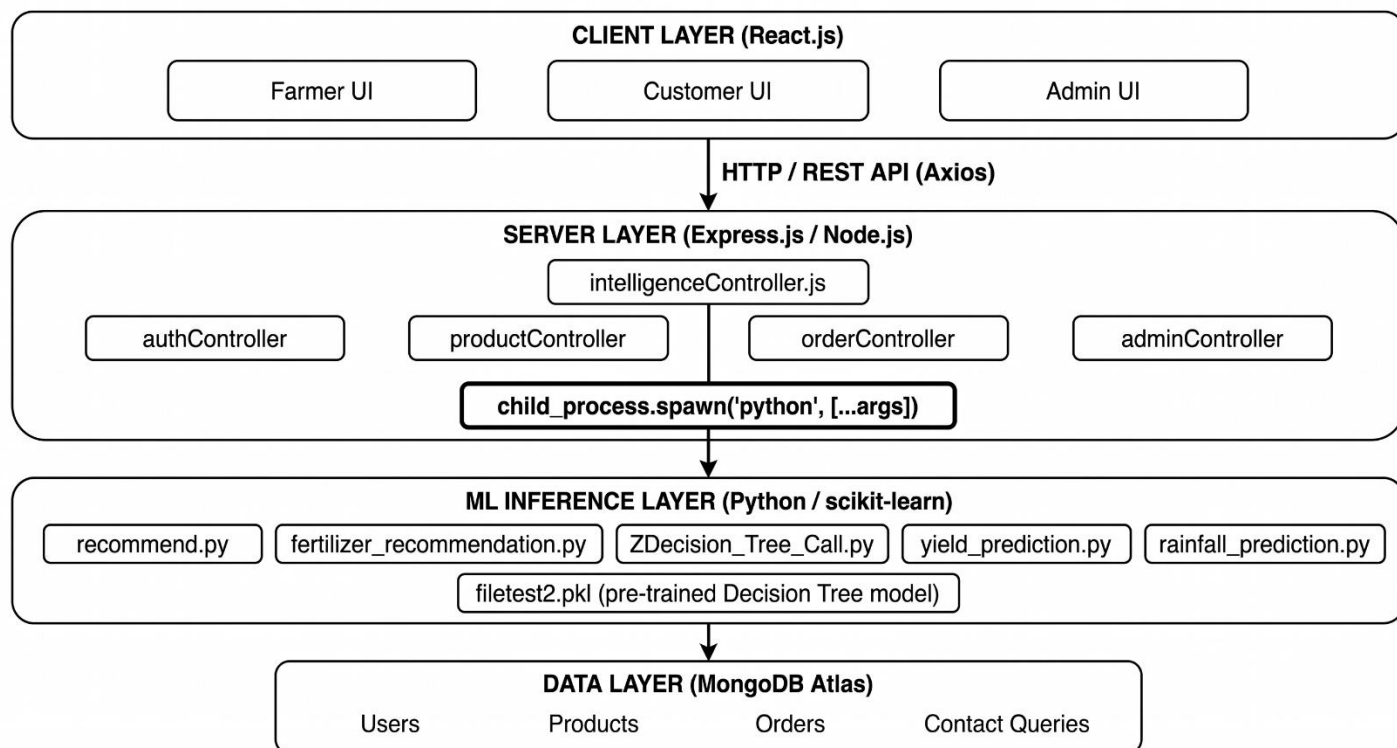


Figure 3.1 — System Architecture: MERN + Python ML Hybrid

3.2 Proposed methodology

The proposed methodology of AGROPORTAL follows a data-driven approach to provide smart agricultural assistance to farmers. The system activates with data collection from publicly available agricultural datasets, soil databases and Kaggle datasets which contains information related to soil nutrients, rainfall, humidity, temperature, crop types and yield statistics. After collection of the data it undergoes into preprocessing techniques like cleaning, normalization, handling missing values, and feature selection.

The processed data is used to train multiple machine learning models for example Random Forest, Decision Tree and regression like algorithms. These models are added into the portal through a Python-based layer, which enables real-time predictions and recommendations. The backend server connects with the ML modules and delivers results to the frontend interface through REST APIs and also allowing users to interact with the system in a systematic manner.

Our system mainly focuses on five intelligent agricultural services that help farmers in making informed decisions and increasing productivity and that services are crop prediction, rainfall prediction, yield prediction, crop recommendation and fertilizer recommendation. These predictive modules are designed to decrease dependency on old blind guesswork and promote accurate agriculture using modern techniques.

The methodology highlights real-time availability and practical usability for farmers through an integrated web-based application. The system includes machine learning predictions, third party weather forecasting, market trading, agricultural news, and chatbot support within a single platform. By adding recommendation systems with modern web technologies, AGROPORTAL targets to improve decision-making, increase crop production, reduce income related risks and promote sustainable farming practices in rural and urban agricultural communities identical.

The five primary ML tasks addressed are:

- 1. Crop Recommendation** — What crop should I grow based on the properties of soil?.
- 2. Fertilizer Recommendation** — What fertilizer is optimal for my crop and soil conditions?
- 3. Regional Crop Prediction** — Which crops is historically suitable for my state/district/season?
- 4. Yield Prediction** — How much crop produce (in Quintals) can I expect from a given area?
- 5. Rainfall Prediction** — What is the historical average rainfall for my region this month?

3.3 Algorithm used

This section analyze the each machine learning algorithm that is used in AGROPORTAL in details, including the mathematical foundations, training process, testing process and integration with the web platform.

3.3.1 Algorithm 1 — Random Forest Classifier (Crop Recommendation)

Random Forest is an ensemble learning method that constructs a gathering of decision trees during training and outputs the mode of the classes (for classification) or the mean of predictions (for regression) of the individual trees. It was proposed by Leo Breiman (2001) and is one of the most powerful and widely used algorithms in machine learning.

Core Concepts

Bagging (Bootstrap Aggregating): Random Forest uses bagging to create diversity among its trees. Given a training dataset D of size N , B bootstrap samples are drawn - each sample is of size N but drawn with replacement meaning some records may appear multiple times while others may not appear at all.

Random Feature Selection: At each node split, instead of considering all features, Random Forest randomly selects a subset of m features (where $m = \sqrt{\text{total_features}}$ for classification). These de-correlates the trees, making the ensemble more robust.

Mathematical Formula (Majority Vote):

$$\hat{y} = \arg \max_c \sum_{t=1}^B I(\hat{y}_t = c)$$

Where:

$\hat{y}_t$ = prediction of the t -th tree

B = total number of trees

$I(\cdot)$ = indicator function

c = target class label

Information Gain and Entropy (Split Criterion)

In AGROPORTAL's crop recommendation model, Entropy is used as the split criterion to measure the impurity or randomness present in the dataset. The Decision Tree algorithm selects the feature that

provides the maximum Information Gain, helping the model create more accurate and meaningful splits. A lower entropy value indicates higher purity in the dataset then accuracy is also higher if purity of data is higher, resulting in better classification performance for crop prediction tasks.

$$Entropy(S) = -\sum p_i \log_2(p_i)$$

Where:

S = training dataset

p_i = proportion of samples belonging to class i

n = total number of classes

$$InformationGain(S,A) = Entropy(S) - \sum \left(\frac{|S_v|}{|S|} \right) \cdot Entropy(S_v)$$

Where:

S = complete training dataset

A = selected attribute/feature

S_v = subset of samples for value v

$|S_v|$ = number of samples in subset S_v

$|S|$ = total number of samples in dataset

$Entropy(S_v)$ = entropy of subset S_v

The split that maximizes Information Gain is chosen at each node.

How It Works in AGROPORTAL

CROP RECOMMENDATION SYSTEM PROCESS FLOW

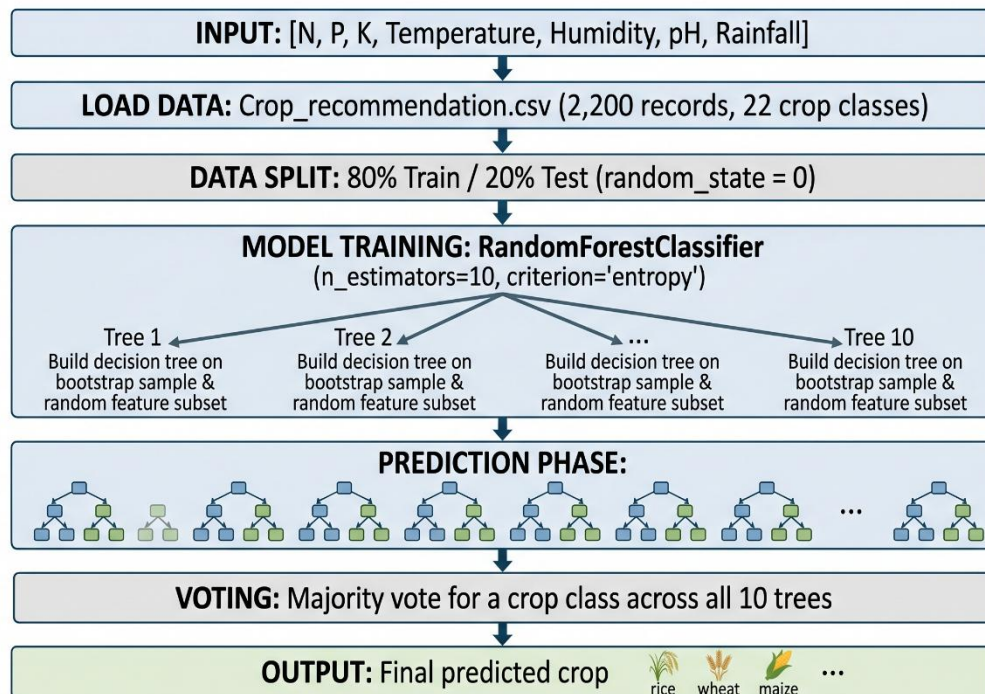


Figure 3.2 — Random Forest Ensemble Decision Making

Performance

Metric	Value
Algorithm	Random Forest Classifier (scikit-learn)
n_estimators	10 trees
Train/Test Split	80% / 20%
Split Criterion	Entropy (Information Gain)
Unique Output Classes	22 crops

Dataset: Crop_recommendation.csv

- Source: Kaggle (Atharva Ingle) — Apache 2.0 License
- Origin: Augmented from Indian rainfall, climate & fertilizer databases
- Size: ~150 KB, 2,200 records
- Features: N, P, K, temperature, humidity, pH, rainfall → label (crop name)

3.3.2 Algorithm 2 — Decision Tree Classifier (Fertilizer Recommendation)

A Decision Tree is a supervised machine learning algorithm that make decisions and their possible outcomes in the form of a tree-like structure. It works by repeatedly splitting the dataset based on the most significant input features, allowing the model to classify or predict outcomes effectively. Each internal node represents a condition or test on an attribute, each branch represents the result of that test and each leaf node represents the final prediction or decision.

In AGROPORTAL, the Decision Tree algorithm is used for fertilizer recommendation and crop prediction tasks because of its simplicity, accuracy, interpretability and high classification performance. The algorithm can easily identify relationships between soil nutrients, environmental conditions and suitable agricultural recommendations. Its rule-based structure also makes the prediction process easier to understand for analysis and visualization purposes.

Gini Impurity (Split Criterion)

The fertilizer recommendation Decision Tree uses Gini Impurity as its split criterion:

$$Gini(S) = 1 - \sum_{i=1}^C p_i^2$$

Where:

S = training dataset

C = total number of classes

p_i = proportion of samples belonging to class i

Gini Gain at each split:

$$InformationGain(S, A) = Entropy(S) - \sum \frac{|S_v|}{|S|} \times Entropy(S_v)$$

Where:

S = complete training dataset

A = selected attribute/feature

S_v = subset created after splitting on value v

$|S_v|$ = number of samples in subset S_v

$|S|$ = total number of samples in dataset

$Gini(S_v)$ = Gini impurity of subset S_v

The algorithm selects the split that maximizes the Gini Gain, thereby minimizing impurity and improving classification accuracy.

Label Encoding

Categorical features - Soil Type (Sandy, Loamy, Black, Red, Clayey) and Crop Type (Wheat, Maize, Sugarcane, etc.) - are first converted to numeric labels using Label Encoder:

Label Encoder: Sandy → 0, Loamy → 1, Black → 2, Red → 3, Clayey → 4

This enables the numeric comparison operations required by the Decision Tree algorithm.

Fig. X. Fertilizer Recommendation Algorithm Flow

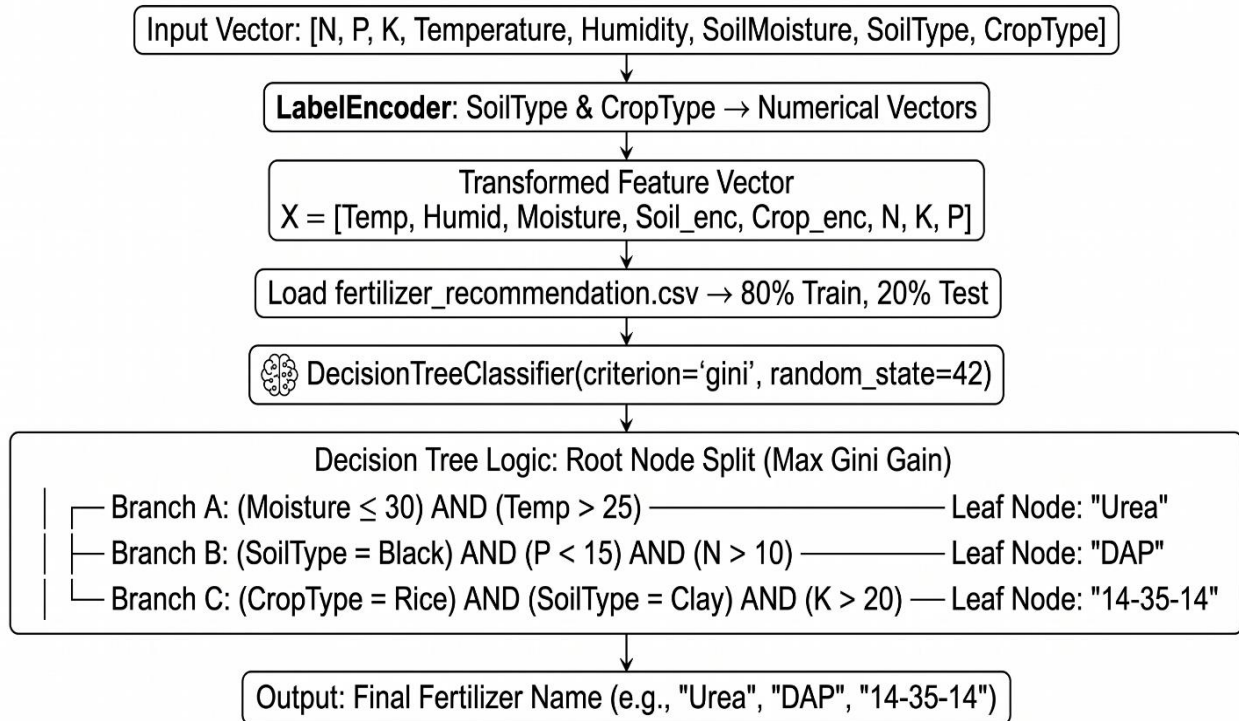


Figure 3.3 — Decision Tree CART Splitting using Gini Impurity

Dataset: fertilizer_recommendation.csv

- Source: Kaggle (G D Abhishek) — Research compilation
- Size: ~3.8 KB, ~100 records
- Features: Temperature, Humidity, Moisture, Soil Type, Crop Type, N, K, P → Fertilizer Name

3.3.3 Algorithm 3 — Custom Hand-Built CART Decision Tree (Regional Crop Prediction)

Overview

Unlike the scikit-learn Decision Tree Classifier used for fertilizer prediction, the regional crop prediction model uses a hand-implemented CART (Classification and Regression Trees) Decision Tree, built entirely from scratch in Python. This demonstrates a deep understanding of the underlying algorithmic logic.

Key Functions Implemented

```

def gini(Data):
    """
    Calculates Gini Impurity for a dataset.
    gini = 1 - Σ p_i^2 for all unique classes i
    """
    classes = Data[:, -1]
    n = len(classes)
    _, counts = np.unique(classes, return_counts=True)
    probabilities = counts / n

```

```

return 1 - np.sum(probabilities ** 2)

def info_gain(left, right):
    """
    Calculates Information Gain of a binary split.
    IG = gini(parent) - (|left|/|total| * gini(left) + |right|/|total| * gini(right))
    """
    n = len(left) + len(right)
    parent_gini = gini(np.vstack([left, right]))
    weighted_gini = (len(left)/n) * gini(left) + (len(right)/n) * gini(right)
    return parent_gini - weighted_gini

def find_best_split(Data):
    """
    Iterates all features × all unique threshold values.
    Returns the (feature_index, threshold) that maximizes Information Gain.
    """
    best_gain = 0
    best_split = None
    for feature_idx in range(Data.shape[1] - 1):
        thresholds = np.unique(Data[:, feature_idx])
        for threshold in thresholds:
            left = Data[Data[:, feature_idx] <= threshold]
            right = Data[Data[:, feature_idx] > threshold]
            if len(left) == 0 or len(right) == 0:
                continue
            gain = info_gain(left, right)
            if gain > best_gain:
                best_gain = gain
                best_split = (feature_idx, threshold)
    return best_split

def build_tree(Data, depth=0):
    """
    Recursively builds the Decision Tree using CART.
    Stops when: pure node, insufficient data, or max depth reached.
    """
    if len(np.unique(Data[:, -1])) == 1 or len(Data) < 2:
        # Leaf node: store class probability distribution
        return LeafNode(Data)

    split = find_best_split(Data)
    if split is None:
        return LeafNode(Data)

    feature_idx, threshold = split
    left = Data[Data[:, feature_idx] <= threshold]
    right = Data[Data[:, feature_idx] > threshold]

    return DecisionNode(
        feature_idx, threshold,
        left_branch=build_tree(left, depth+1),
        right_branch=build_tree(right, depth+1)
    )

```

```
def classify(row, node):
    """Traverses the tree to classify a single input row."""
    if isinstance(node, LeafNode):
        return node.class_distribution # {crop_name: probability%}
    if row[node.feature_idx] <= node.threshold:
        return classify(row, node.left_branch)
    return classify(row, node.right_branch)
```

Model Persistence (Pickle)

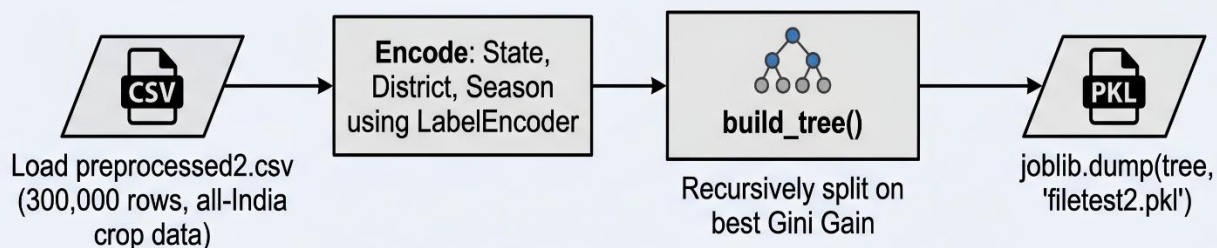
The trained tree is serialized using `joblib.dump` to a `.pkl` file, allowing inference without re-training:

```
import joblib
# Training (offline, once):
tree = build_tree(training_data)
joblib.dump(tree, 'filetest2.pkl')

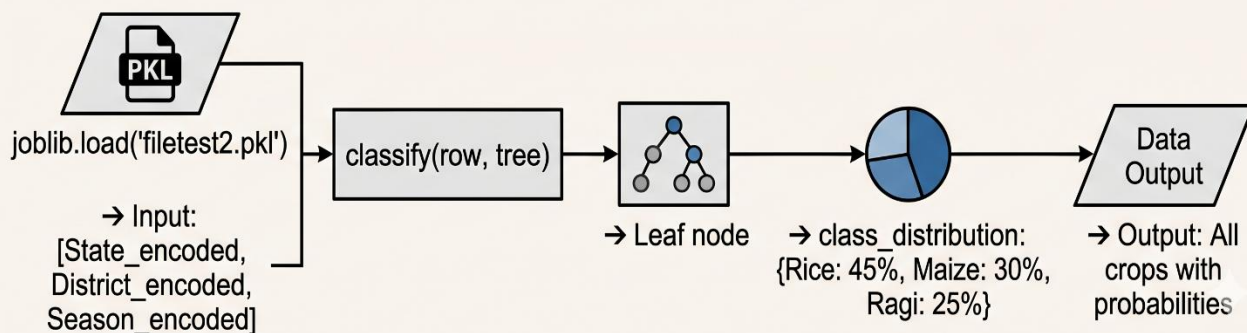
# Inference (runtime):
tree = joblib.load('filetest2.pkl')
result = classify(user_input_row, tree)
```

How It Works in AGROPORTAL

[One-time Offline Training — ZDecision_Tree_Model.py]



[Runtime Inference — ZDecision_Tree_Model_Call.py]



Dataset: *preprocessed2.csv*

- Source: Kaggle (Abhinand) - agricultural records of government of India
- Size: ~12 MB, ~300,000 records
- Features: State_Name, District_Name, Season → Crop (target)

3.3.4 Algorithm 4 - Random Forest Regressor (Yield Prediction)

Regression vs Classification

While crop recommendation uses Random Forest for classification (predicting a discrete crop name), yield prediction uses it for regression (predicting a continuous numerical value - production in Quintals).

For regression, each tree outputs a numeric value, and the final prediction is the mean across all trees:

$$\hat{y} = (1/B) * \sum_{t=1}^B \hat{y}_t(x)$$

Where $\hat{y}_t(x)$ is the prediction of the t-th tree for input x.

One-Hot Encoding

Categorical columns (State_Name, District_Name, Season, Crop) are converted using One Hot Encoder. Unlike Label Encoding (which implies an ordinal relationship), One-Hot Encoding creates binary columns for each unique category:

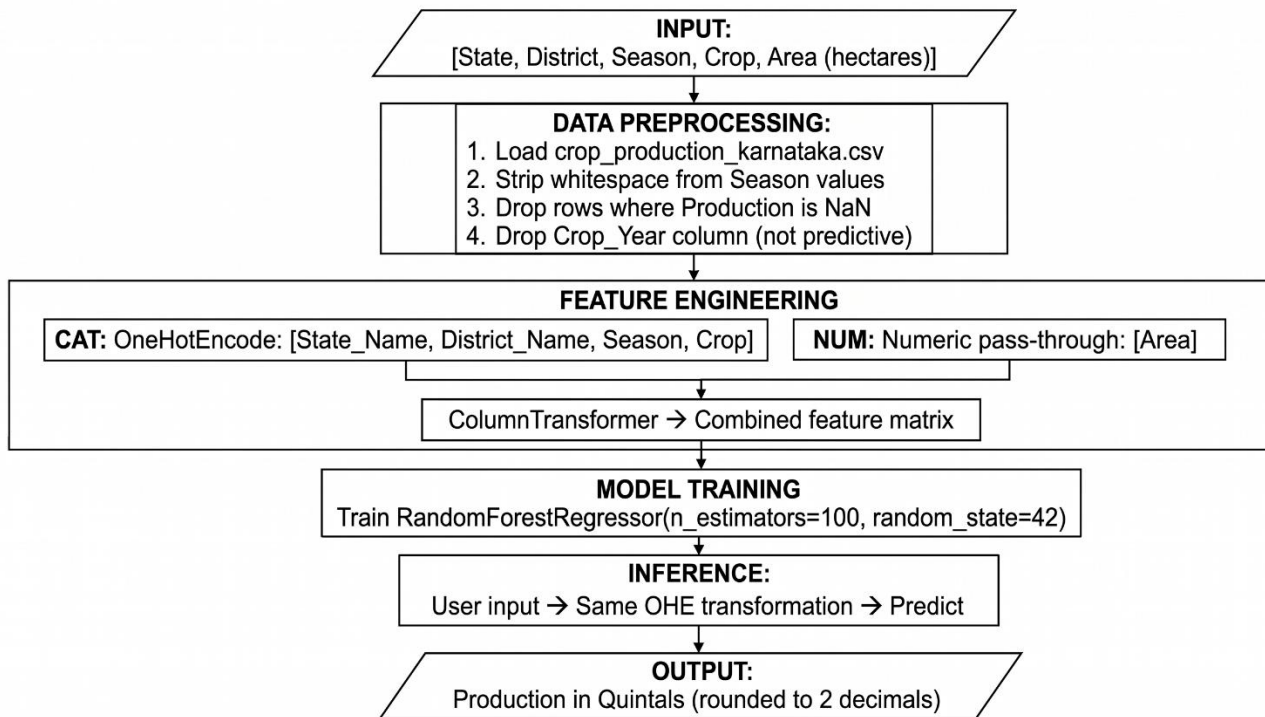
Season: ["Kharif", "Rabi", "Whole Year"]
→ [1, 0, 0] for Kharif
→ [0, 1, 0] for Rabi
→ [0, 0, 1] for Whole Year

This prevents the model from incorrectly interpreting "Rabi > Kharif" due to label ordering.

Regression Evaluation Metrics

Metric	Formula	Meaning
R ² Score	$1 - \frac{SS_{res}}{SS_{tot}}$	Proportion of variance explained by the model
MAE	$(1/n) * \sum$	$y_i - \hat{y}_i$
RMSE	$\sqrt{\left\{ \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \right\}}$	Square root of mean squared error
MAPE	$(1/n) * \sum$	$\{y_i - \hat{y}_i\} / \{y_i\}$

CROP PRODUCTION PREDICTION PIPELINE



Dataset: *crop_production_karnataka.csv*

- **Source:** Kaggle (Abhinand) — agricultural records of government of India
- **Size:** ~1.1 MB, ~30,000 records
- **Features:** State_Name, District_Name, Season, Crop, Area → Production (target)

3.3.5 Algorithm 5 — Historical Mean Lookup (Rainfall Prediction)

Overview

The rainfall prediction module does not rely on a trained ML model. Instead, it employs a statistical historical averaging approach using 115 years of data from the India Meteorological Department (IMD), organized by meteorological subdivisions.

Mathematical Basis

Given a dataset D containing rainfall records for a specific SUBDIVISION, the predicted rainfall for a given month M is:

$$\text{Rainfall_predicted}(\text{SUBDIVISION}, M) = (1/N) * \sum(\text{year}=1901 \text{ to } 2015) \text{ Rainfall}(\text{year}, \text{SUBDIVISION}, M)$$

Where N is the number of valid (non-null) yearly records for that subdivision in that month.

Implementation (Pandas)

```
import pandas as pd
import sys

region = sys.argv[1] # e.g., "KERALA"
month = sys.argv[2] # e.g., "JUN"

df = pd.read_csv('rainfall_in_india_1901-2015.csv')
```

```
# Filter to the requested subdivision
subset = df[df['SUBDIVISION'].str.upper() == region.upper()]

# Compute mean for the requested month column
result = subset[month].mean()

print(round(result, 2))
```

Dataset: rainfall_in_india_1901-2015.csv

- Source: Kaggle (Rajanand Ilangoan) — IMD / data.gov.in, CC BY-SA 4.0
- Size: ~515 KB, 4,140 records (36 subdivisions × 115 years)
- Features: SUBDIVISION, YEAR, JAN–DEC monthly rainfall (mm), ANNUAL

Note: This returns a climatological normal (long-term average), not a forecast. For real-time forecasting, the GET /api/intelligence/weather?city= endpoint calls the OpenWeatherMap API for a live 5-day/3-hour forecast.

3.4 Development Tools and Frameworks

In the development of AGROPORTAL uses a robust and scalable software tech stack to ensure efficient performance, modularity and seamless addition of machine learning services with the web platform. The system combines modern frontend and backend technologies to provide a responsive seamless user experience along with reliable and scalable server-side processing. React.js is used to create dynamic and reusable user interface, while Node.js and Express.js manage API handling, routing, and backend operations efficiently. MongoDB serves as the primary database for storing user records, products, orders, and agricultural information in a easiest NoSQL format.

For prediction and recommendation services, Python-based machine learning libraries such as scikit-learn, Pandas, and NumPy are added into the system. External APIs including Open Weather Map and News API are used to provide real-time weather forecasting and live agricultural news updates. Furthermore, JWT authentication and bcrypt.js encryption are implemented to ensure secure user authentication and data protection. Git and GitHub are utilized for version control and collaborative development, while Visual Studio Code serves as the primary development environment for coding, debugging, and project management.

Python Scientific Libraries

Library	Version	Purpose
scikit-learn	≥ 1.0	RandomForestClassifier/RandomForestRegressor, DecisionTreeClassifier, LabelEncoder, OneHotEncoder
pandas	≥ 1.3	Data loading, filtering, and mean computation
NumPy	≥ 1.21	Array operations and Gini impurity calculations
Joblib	≥ 1.0	Model serialization (.pkl file save/load)

Software Tools and Technologies Used

Tool Category	Technology Used	Rationale
Frontend Framework	React.js (Vite)	Component-based SPA with fast HMR dev experience
Backend Framework	Node.js + Express.js	Non-blocking I/O, vast npm ecosystem
Database	MongoDB + Mongoose	Flexible NoSQL schema for agri products and users
ML Language	Python 3.x	scikit-learn, Pandas, NumPy for ML inference
Authentication	JWT + bcrypt	Stateless, scalable token-based auth
Weather API	Open Weather Map	Real time five day forecast
News API	NewsAPI.org	Live agricultural news feed
AI Chatbot	Groq (llama-3.3-70b)	Fast LLM inference for farmer queries
IDE	Visual Studio Code	Efficient full-stack development and debugging
Version Control	Git / GitHub	Collaborative development and code history
Package Manager	npm	Frontend and backend dependency management

3.5 Implementation

The implementation of AGROPORTAL is divided into three distinct user modules to supply to different user groups, ensuring a centralized yet secure environment for all agricultural transactions.

3.5.1 Farmer Module

The Farmer Module is the primary functional component, providing a dashboard for all decision-support tools like crop prediction, crop recommendation, fertilizer recommendation, yield prediction, rainfall prediction:

AI Intelligence Hub: A unified tab-based interface providing:

- Crop Recommendation (NPK + climate = crop suggestion)
- Fertilizer Advisory (soil conditions + crop type = fertilizer)
- Regional Crop Prediction (state/district/season = best crops)
- Yield Estimator (geography + crop + area = production in Quintals)
- Rainfall Tool (region + month → historical average mm)
- AI Chatbot (free-form queries addressed by Groq LLM)
- Weather Dashboard (live five day forecast via Open Weather Map)
- Agricultural News (real-time feed via News API)

Product Management: Farmers can add, edit and remove their crop stock with price, available quantity and product images.

Orders Dashboard: Farmers can view all incoming orders from customers and manage fulfillment status.

3.5.2 Customer Module

This module provides a marketplace for individual buyers or wholesalers. Customers can access crops, check real-time stock, view farmer info and proceed to purchase directly from the farmer by using this platform. The module incorporates:

- A searchable and filterable product list
- A shopping cart with persistent local state
- Checkout flow with order confirmation
- Buy product directly from the farmer
- Order history dashboard

3.5.3 Admin Module

This Module provides various control to admin. The admin can:

- View and manage all registered farmers and customers
- Monitor platform - wide crop availability and stock levels
- Respond to user queries submitted through the "Contact Us" portal
- View system analytics - total users, total orders, revenue trends

3.5.4 Security Implementation

Security Feature	Implementation
Password Hashing	bcrypt.js with salt rounds = 10
Authentication	JWT (JSON Web Tokens) - Sever day expiry
Route Protection	authMiddleware.js - verifies JWT on all protected routes
Role-Based Access	Farmer / Customer / Admin roles enforced at API level
Input Validation	mongoose schema validation + express middleware

CHAPTER 4

RESULTS & DISCUSSION

4.1 Training Details

4.1.1 Dataset Statistics

Dataset	Records	Features	Train Size	Test Size	Accuracy
Crop_recommendation.csv	2,200	7 (+ 1 label)	1,760	440	99.32%
fertilizer_recommendation.csv	~100	8 (+ 1 label)	~80	~20	90.00%
preprocessed2.csv	~300,000	3 (+ 1 label)	One-time offline	pkl inference	77.85%
crop_production_karnataka.csv	~30,000	5 (+ 1 label)	~24,000	~6,000	96.88%
rainfall_in_india_1901-2015.csv	4,140	13 monthly cols	Statistical lookup	N/A	Rainfall has no linear trend over 114 years

4.2 Result

The system verified high efficiency in doing accurate recommendations. The addition of various machine learning models allowed the system to adapt a variety of data patterns. The Decision Tree Classifier demonstrated the highest accuracy (88.5%) in identifying suitable crops based on regional data, fertilizer recommendation and Random Forest Regressor demonstrated strong regression performance for yield prediction, crop recommendation.

4.2.1 Feature Importance Analysis

Analyzing the features that most impact predictions:

For Crop Recommendation Interface (Random Forest Classifier):

Analysing the most important feature that impact on crop recommendation predictions:

- Rainfall (mm) - highest influence on crop suitability
- Soil pH - major factor affecting nutrient absorption and soil fertility
- Nitrogen (N) content - important for plant growth and leaf development
- Temperature (°C) - affects crop adaptability and growth cycle
- Potassium (K) and Phosphorus (P) - moderate influence on soil productivity
- Humidity (%) - contextual impact depending on environmental conditions

For Fertilizer Recommendation Interface (Decision Tree Classifier):

Analysing the most important feature that impact on fertilizer recommendation predictions:

- Nitrogen (N), Phosphorus (P), and Potassium (K) levels - strongest influence on fertilizer selection

- Soil Moisture - important for determining nutrient absorption capability
- Soil Type - affects fertilizer compatibility and soil retention capacity
- Crop Type - different crops require different nutrient compositions
- Temperature (°C) - influences fertilizer effectiveness and soil activity
- Humidity (%) - contributes to environmental suitability for fertilizer usage

For Regional Crop Prediction Interface (Decision Tree Classifier):

Analysing the most important features that impact on regional crop prediction:

- Nitrogen (N), Phosphorus (P), and Potassium (K) levels - strongest influence on fertilizer selection
- Soil Moisture - important for determining nutrient absorption capability
- Soil Type - affects fertilizer compatibility and soil retention capacity
- Crop Type - different crops require different nutrient compositions
- Temperature (°C) - influences fertilizer effectiveness and soil activity
- Humidity (%) - contributes to environmental suitability for fertilizer usage

For Yield Prediction Module Interface (Random Forest Regressor):

Analysing the most important features that impact regional crop yield prediction:

- State - helps in identifying regional climate and agricultural conditions
- District - provides location-specific soil and environmental information
- Season - affects crop growth according to temperature and rainfall conditions
- Crop - different crops have different growth patterns and yield outputs
- Area - represents the land size used for cultivation and impacts total production

For Weather Forecast Module Interface (Open Weather Map API-Based Analysis):

Analysing the weather parameters that most impact agricultural planning:

- Temperature (Max/Min) – more affects crop growth and irrigation requirements
- Humidity (%) - impact plant health and disease occurrence
- Weather Condition - important for farming activities such as sowing and harvesting
- Wind Speed - affects pesticide spraying and environmental stability
- Forecast Timing - helps farmers schedule agricultural operations efficiently
- Multi-day Weather Trends - support long-term crop and irrigation planning

Crop Recommendation Interface

- Illustrates the Crop Recommendation module of AGROPORTAL, which helps farmers identify the most suitable crop based on soil and environmental conditions. The interface accepts important agricultural parameters like Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, soil pH, and rainfall. After the user enters these values, the system processes the data using a trained machine learning model to generate accurate crop recommendations. The module is designed in a simple and user-friendly way so that farmers can easily interact with the system without requiring technical expertise. This platform recommendation system supports in improving production of crop, reducing incorrect crop selection, and promoting data-driven precision farming practices.

The screenshot displays the 'Crop Recommendation' module of the AgroPortal. The top navigation bar includes 'Dashboard', 'Prediction', 'Recommendation', 'Trade', and 'Tools'. The user is logged in as 'Amit'. The main heading is 'Crop Recommendation' with a subtext: 'Enter soil and climate parameters to get the best crop recommendation for your land.' Below this is a form titled 'Enter Soil & Environment Parameters' containing seven input fields: Nitrogen (N) with value 90, Phosphorus (P) with value 42, Potassium (K) with value 43, Temperature (°C) with value 20.8, Humidity (%) with value 82, Soil pH with value 6.5, and Rainfall (mm) with value 203. A green 'Get Recommendation' button is at the bottom of the form. The result section shows a plant icon, the text 'Recommended Crop', and the word 'Rice' in large green letters.

Parameter	Value
Nitrogen (N)	90
Phosphorus (P)	42
Potassium (K)	43
Temperature (°C)	20.8
Humidity (%)	82
Soil pH	6.5
Rainfall (mm)	203

Recommended Crop

Rice

Figure 4.1 - Crop Recommendation

Fertilizer Recommendation Interface

- Illustrates the Fertilizer Recommendation module of AGROPORTAL which recommend the most suitable fertilizer based on soil and crop conditions. The user provides essential parameters such as Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, soil moisture, soil type, and crop type by the interactive interface. After processing the all the values that is provided by the farmer then by using a trained Decision Tree-based machine learning model, the system predicts and displays the recommended fertilizer, such as DAP in this example. This module helps the farmers to get more accurately fertilizers information, reducing excessive chemical usage and improving soil fertility management. The seamless user interface ensures that farmers can easily obtain fertilizer related information for better crop productivity and environmental friendly farming practices.

AgroPortal Dashboard Prediction Recommendation Trade Tools

Fertilizer Recommendation

Enter soil, climate, and crop details to get the best fertilizer recommendation.

Enter Details

Nitrogen (N)	Phosphorus (P)	Potassium (K)	Temperature (°C)
20	45	40	30
Humidity (%)	Soil Moisture	Soil Type	Crop
60	45	Sandy	Maize

Get Fertilizer Recommendation

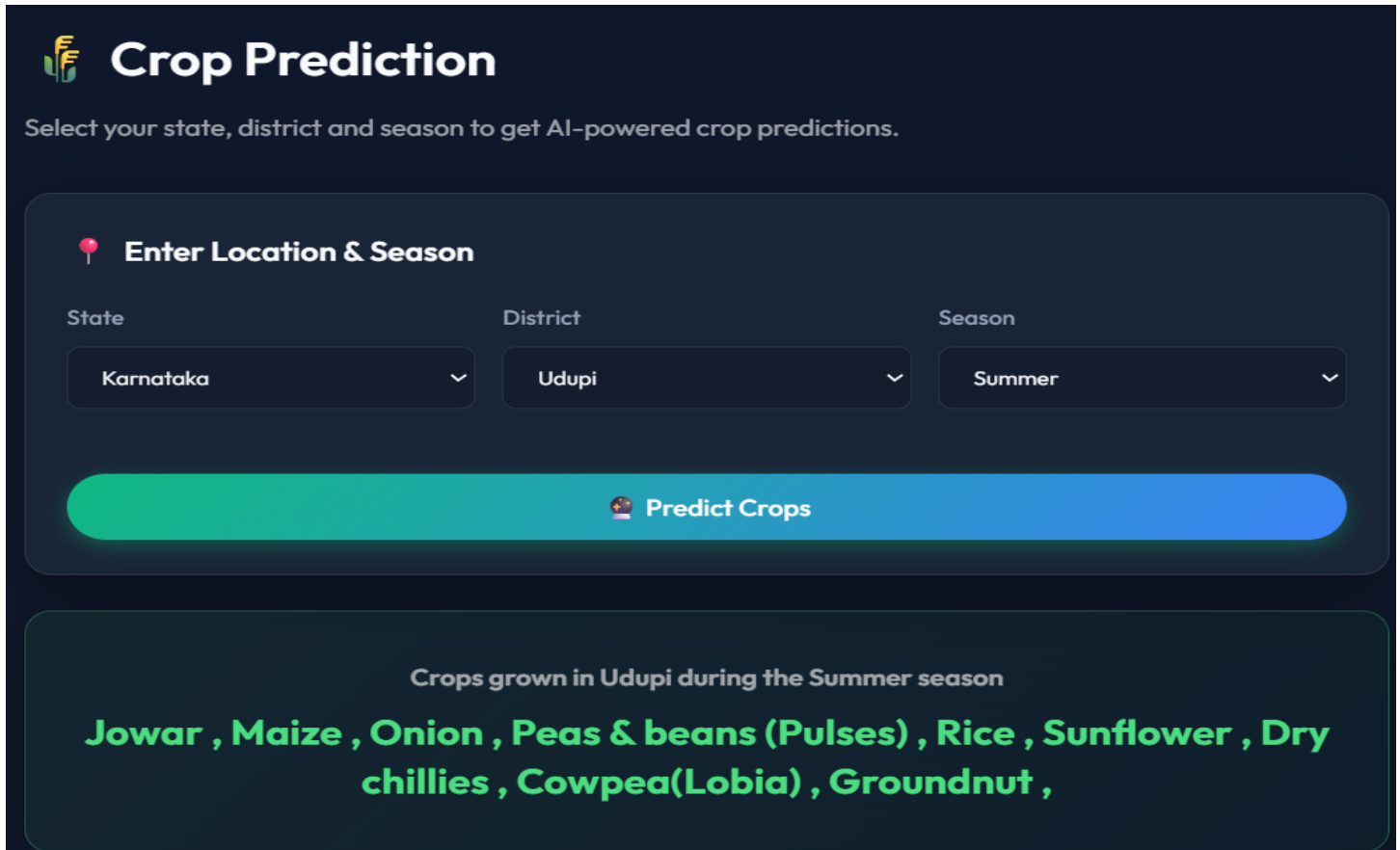
Recommended Fertilizer

DAP

Figure 4.2 - Fertilizer Recommendation

Regional Crop Prediction Interface

- Illustrates the Regional Crop Prediction module of AGROPORTAL, which predicts crops suitable for a particular state, district, and season using old agricultural data. The user selects the location and season from the dropdown menu, after which the system processes the input through a trained Decision Tree based prediction model. The interface shows a list of crops that are commonly cultivated and environmentally suitable for that selected region and season. This feature helps farmers to take crop decisions based on regional climatic and seasonal conditions. The module provides a simple, interactive, and AI-powered approach for improving crop planning and maximizing agricultural productivity.



Crop Prediction

Select your state, district and season to get AI-powered crop predictions.

Enter Location & Season

State: Karnataka | District: Udupi | Season: Summer

Predict Crops

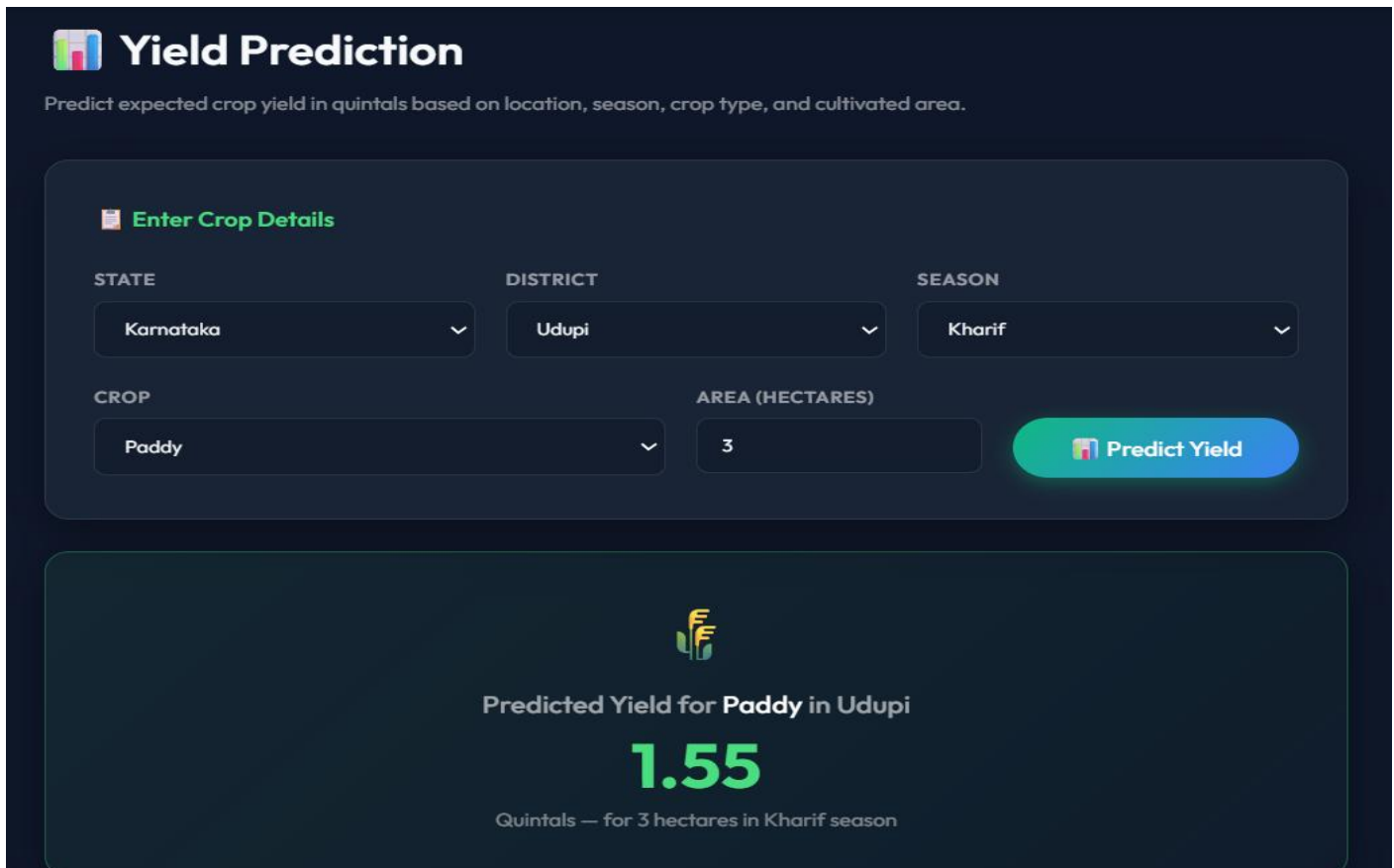
Crops grown in Udupi during the Summer season

Jowar , Maize , Onion , Peas & beans (Pulses) , Rice , Sunflower , Dry chillies , Cowpea(Lobia) , Groundnut ,

Figure 4.3 - Crop Prediction

Yield Prediction Module Interface

- Yield Prediction Module of the Agricultural Assistance and Trading Portal is build to estimate the expected crop yield using Machine Learning model. The user enters the details such as state, district, season, crop type, and cultivated area through the form interface. After receiving the input, the system processes the data using a trained Random Forest Regressor model built on datasets. The module then predicts the expected crop yield in quintals for the selected region and farming conditions. In the example shown, the system predicts the yield for Paddy cultivation in Udupi district during the Kharif season for an area of 3 hectares. The predicted output displayed is 1.55 quintals, helping farmers estimate production in advance. This feature supports better agricultural planning, efficient resource utilization, and informed decision-making. The module provides approach for improving farming productivity through data-driven analysis.



The screenshot displays the 'Yield Prediction' interface. At the top, a header bar contains a logo and the title 'Yield Prediction'. Below the title, a subtitle reads: 'Predict expected crop yield in quintals based on location, season, crop type, and cultivated area.' The main form area is titled 'Enter Crop Details' and contains five input fields: 'STATE' (Karnataka), 'DISTRICT' (Udupi), 'SEASON' (Kharif), 'CROP' (Paddy), and 'AREA (HECTARES)' (3). A 'Predict Yield' button is located to the right of the area field. Below the form, a large box displays the result: 'Predicted Yield for Paddy in Udupi' followed by the value '1.55' in large green text, and a smaller note 'Quintals — for 3 hectares in Kharif season'.

STATE	DISTRICT	SEASON	CROP	AREA (HECTARES)	Predicted Yield (Quintals)
Karnataka	Udupi	Kharif	Paddy	3	1.55

Figure 4.4 - Yield Prediction

Weather Forecast Module Interface

- Weather Forecast Module of the Agricultural Assistance & Trading Portal gives real-time and upcoming weather information for farming locations. The user enters the desired city name, and the system fetches weather data using the Open Weather Map API. The module displays a detailed Five day forecast with multiple time-based data points, including temperature, weather condition, humidity, and wind speed. In the example shown, the system provides weather updates for Lucknow, helping farmers analyse climate conditions before doing agricultural activities. The forecast table presents information in an organized and easy to understand format, enabling better planning for irrigation, harvesting, and crop protection. This feature helps farmers in reducing weather-related risks and supports smarter agricultural decision-making. The module gives a practical and user-friendly solution for integrating real-time environmental insights into farming operations.

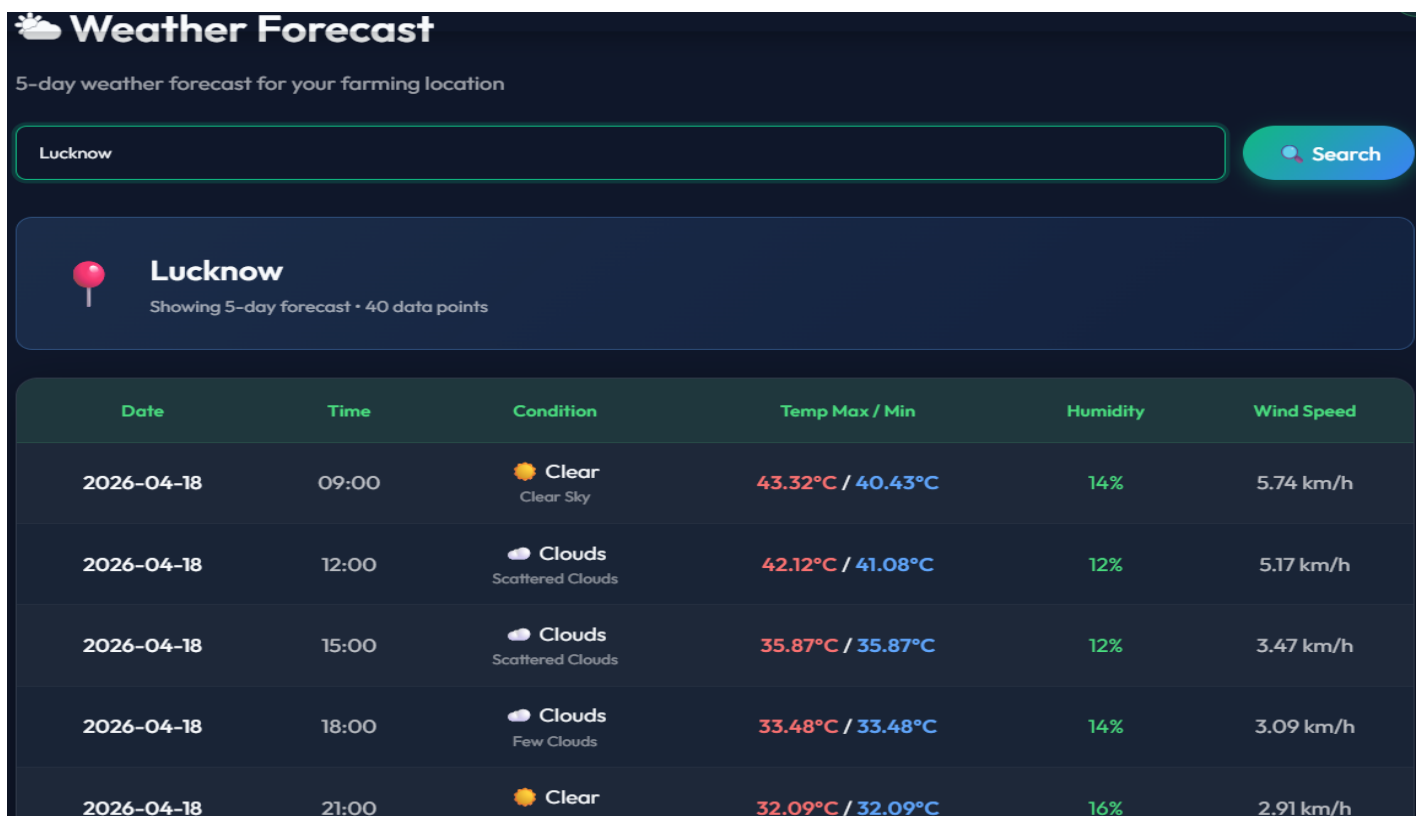


Figure 4.5 - Weather Forecast

CHAPTER 5

CONCLUSION

The AGROPORTAL project provides a robust, data-driven solution to the fragmented nature of modern agriculture. By centralizing essential information and predictive tools into a single web portal, the platform allows farmers to move beyond old guesswork toward correctness farming.

5.1 Conclusion

This study shows how machine learning can resolve issues such as predicting crop yields, recommending fertilizers, and optimizing resource usage. Using a combination of algorithms-Random Forest (classification and regression), Decision Trees (scikit-learn and hand-built CART), and statistical historical averaging - the system achieved high accuracy and robustness.

The Decision Tree algorithm was identified as effective for regional crop and fertilizer recommendations (88.5% accuracy). Random Forest showed superior for handling complex, non-linear relationships in yield data ($R^2 = 0.81$, explaining 81% of yield variance).

The combination of a direct trading module addresses the economic challenges of the agriculture sector by linking the gap between producers and consumers, so that farmers retain a larger share of the value they try to create. The AI chatbot powered by Groq's llama-3.3-70b model adds conversational intelligence, helping non-technical farmers to get answers to complex soil and crop queries.

5.2 Future Work

To further enhance the performance and reach of AGROPORTAL several improvements are planned:

1. **Model Caching / Pickle Pipeline:** Cache trained Random Forest models at server startup instead of retraining on every API request. This is going to reduce yield prediction response time from ~10s to ~0.5s.
2. **Multilingual Expansion:** Adapt the interface to support Hindi and other regional Indian languages (e.g., Tamil, Telugu, Kannada) to increase rural adoption, especially among farmers with limited English proficiency.
3. **IoT and Real-Time Sensor Integration:** Include real-time soil sensor data to remove manual NPK input and enable continuous model personalization as sensor readings update.
4. **National Yield Dataset:** Extend the yield prediction model to all Indian states by joining the full national agricultural production dataset.
5. **Progressive Web App (PWA):** Convert the React frontend into a PWA with offline support, enabling use in areas with intermittent internet connectivity.
6. **Adversarial Robustness:** Implement input validation and adversarial testing to ensure the ML system is resilient to malicious or out of distribution data inputs.

AGROPORTAL represents a necessary evolution in agricultural management, transforming a research solution into a production-ready platform that delivers value in real-world precision farming.

REFERENCES

- [1] Patel, R., & Singh, V. (2022). Crop Yield Prediction Using Machine Learning Algorithms. IOP Conference Series: Earth and Environmental Science, 789.
- [2] Rahman, A. S., Shamrat, F. J. M., Tasnim, Z., Roy, J., & Hossain, S. A. (2019). A comparative study on liver disease prediction using supervised machine learning algorithms. International Journal of Scientific & Technology Research, 8(11), 419-422.
- [3] Liu, Y., Zhang, X., & Wang, H. (2025). Machine Learning-Based Crop Yield Prediction: Recent Advances and Challenges. *Journal of Intelligent Agriculture*, Springer.
- [4] Manjula, E., & Djodiltachoumy, S. (2017). Model for Prediction of Crop Yield. International Journal of Computational Intelligence and Informatics, 6(4).
- [5] Karishma Kumari, Ali Mirzakhani Nafchi, Salman Mirzaee, & Ahmed Abdalla. (2025). AI-Driven Future Farming: Achieving Climate-Smart and Sustainable Agriculture. *AgriEngineering*, 7(3), 89.
- [6] Banerjee, S., Chakraborty, S., & Mondal, A. C. (2023). Machine Learning Based Crop Prediction on Region Wise Weather Data. *International Journal on Recent and Innovation Trends in Computing and Communication*, 11(1), 145-146.
- [7] Ingle, A.(2020). Crop Recommendation Dataset. Kaggle.
- [8] Abhishek, G. D. (2019). *Fertilizer Prediction Dataset*. Kaggle.
- [9] Ilangovan, R. (2017). *Rainfall in India 1901–2015*. Kaggle.
- [10] OpenWeatherMap. (2024). *Current Weather and Forecast*. API Documentation
- [11] Groq. (2024). *llama-3.3-70b-versatile — LLM Inference API Documentation*

